SDC

Hackathon · Phase 1

Preliminary Project Proposal

Lumina Nusantara · Domain 1 — The Aiversecity Future-Ready Workforce Domain

SDG Track: SDG 4 — Quality Education (sub-target 4.4, with 8.5)

Team Leader: DU YULIN · Xiamen University Malaysia (XMUM)

Live prototype: https://sdc2026-lumina-nusantara-65cnugvuxdle5dgp9ky3fs.streamlit.app

Code repository: https://github.com/douyinlin/sdc2026-lumina-nusantara

Submission deadline: 25 October 2026

Phase 1 — Preliminary Project Proposal. Answers below follow the official template order and the stated word limits (2.1 ≤ 250 words, 2.2 ≤ 300 words, 3.2 ≤ 200 words).

1. TEAM AND PROJECT OVERVIEW

Team Name	Lumina Nusantara
Selected Domain	Domain 1 — The Aiversecity Future-Ready Workforce Domain
Selected SDG Track (1–17)	SDG 4 — Quality Education (sub-target 4.4, with 8.5)
Project / Solution Name	SkillRadar MY
Name (Leader)	DU YULIN
Email & Phone (Leader)	CYS2609009@xmu.edu.my / +60 11-1798 1756
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Number of Team Members	5

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1 The Problem Context

Malaysia's labour market is being reshaped faster than its training system can respond. 56% of Malaysian professionals already use AI tools at work, yet only 26% have received formal training in them (Hays Asia Salary Guide 2025), and the Ministry of Science, Technology and Innovation expects more than 600,000 workers to need reskilling within three to five years.

The people most exposed are rarely the loudest in this debate: a 34-year-old customer-service supervisor in Klang whose tasks are being automated one by one; a TVET graduate applying for entry roles that no longer exist; an L&D manager in a 200-person SME who must set next year's training budget without evidence of which roles will still be viable.

Three failures repeat. Workers cannot see which skills are rising, so they buy credentials that are already commoditised. Employers cannot translate AI trends into a training plan, so budgets default to generic soft-skills courses. And career guidance is a snapshot: a one-off skills assessment in a market that keeps moving, so the advice expires before the learner finishes the course.

The cost is economic and social — wasted tuition, slower wage growth, and widening gaps for older workers, women returning from care work, and rural learners who cannot reach premium bootcamps. Without a system that updates as AI advances and shows how each role connects to the SDGs, reskilling stays an expensive guess rather than a pathway out.

2.2 The Proposed AI Solution

Intake. A user answers a short questionnaire on role, years of experience, tools and interests, or uploads a CV that we parse into a normalised skill profile. Every skill maps to a shared taxonomy (O*NET / ESCO identifiers) through an alias dictionary, so "Zendesk" and "crm" both resolve to CRM administration.

Trend layer. A scheduled job ingests public signals — WEF Future of Jobs indicators, O*NET skill updates, sampled public job postings and the HRD Corp course catalogue — into a versioned trend snapshot. Each record stores its source URL, retrieval date and metric, and the interface exposes that provenance on every forecast, so anyone can see when the data changed and what moved.

Matching. For each candidate future role we compute an explainable score from four visible components: skill coverage, transferability of adjacent skills, demand trend and gap cost. The user sees the arithmetic and the exact missing skills, not a black box.

Pathway and scenarios. Missing skills are ordered cheapest-first into a week-by-week plan built from free or low-cost resources, each with a link and estimated hours. The ranking runs twice — moderate and fast AI disruption — so the user sees how robust a choice is. Every role and skill carries SDG tags (primary 4.4 and 8.5, with 8.2 and 9.1 where relevant).

From the Phase 1 workshops we apply GenAI APIs for CV parsing and plain-language explanations, a relational schema for profiles, trends and courses, and cloud deployment on Replit and RunPod serverless GPUs, so the prototype is genuinely online rather than a slide.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1 Conceptual Tech Stack

Layer	Technology	Why
UI	Streamlit (Python)	Fast iteration, runs unchanged on Replit and Streamlit Cloud
Service layer	Python modules (src/utils.py)	Testable matching / pathway / SDG logic
Data	SQLite locally, Postgres on hosted deployment	Versioned trend snapshots, profiles, courses
AI	RunPod Serverless + sentence-transformers; optional hosted LLM	GPU embeddings for CV parsing; LLM only phrases explanations
Ingestion	Python scheduled job, JSON snapshots	Each snapshot carries source_url + retrieved_at
Hosting	Replit; Streamlit Community Cloud	Public live demo URL for judges
Tests	python -m unittest discover tests	Regression on score and pathway logic

System block diagram — see the appended diagram (*docs/architecture.png* in the repository).

> User → Streamlit UI → Matching & pathway engine → SQLite/Postgres; Ingestion job → trend_snapshots(source_url, retrieved_at); Streamlit UI ↔ RunPod Serverless (CV parsing, embeddings).

Sponsor-provided GenAI API or Cloud Platform: Yes. The team used the official SDC sponsor-provided resources for development and hosting (see the Technical Tools Declaration, Section 5). The public live demo is deployed on Streamlit Community Cloud with the same code base.

3.2 Design-Thinking & Deployment Strategy

Empathise and define. Five semi-structured interviews — one HR executive, one TVET lecturer, two working adults considering a switch, one final-year student — plus a review of Malaysian reskilling reports. The recurring sentence was "I don't know what to learn next, and I don't trust course adverts". We defined the job to be done as: in under ten minutes, understand which two or three roles are realistic for me and what the first step is.

Ideate and prototype. Paper flows, then a clickable Streamlit prototype tested with the same five people; we cut features that needed data we could not source honestly.

Test and de-risk. The score is transparent by construction, every claim links to a dated source, and the RunPod dependency has a local rule-based fallback so a live demo cannot fail on stage.

Deploy. Replit runs the app and holds secrets; RunPod serves embeddings; a weekly snapshot job keeps trends current. Phase 2 adds the HR view and automatic re-planning.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

> Baselines are taken from published reports or our own five-person interviews. The interview count is stated so judges can weigh the evidence.

Metric Category	Your Response
Target SDG Sub-Target(s)	**4.4** — substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment and entrepreneurship. **8.5** — achieve full and productive employment and decent work for all. Secondary: **4.3** equal access to affordable technical and vocational training.
Current Problem Baseline	56% of Malaysian professionals use AI tools at work but only **26%** have received formal training (Hays Asia Salary Guide 2025); **600,000+** workers need reskilling within 3–5 years (MOSTI). A learner currently spends 3–8 weeks (median 4) on self-directed search across job boards and course catalogues before committing to a route (our five interviews).
Core Measurable KPI	(1) **Minutes from profile to a ranked, sourced reskilling plan** — target ≤5 min, measured in-app. (2) **Top-3 relevance rate** — share of pilot users rating at least one of the three recommended roles as realistic (5-point scale ≥4). (3) **Source coverage** — share of displayed recommendations carrying a source URL and retrieval date; must be 100%.
Phase 2 Impact Target	**70%** top-3 relevance across ≥30 pilot users; ≥**40%** reduction in time-to-plan versus the 4-week baseline; **100%** of role and skill cards traceable to a dated public source; ≥**60%** of pilot users complete step 1 of their pathway within 30 days.

Sources: Hays Asia Salary Guide 2025; Ministry of Science, Technology and Innovation (MOSTI) reskilling estimate as cited in the SDC Domain 1 brief; WEF Future of Jobs Report; O*NET database (U.S. Department of Labor); HRD Corp course directory.

5. ELIGIBILITY & READINESS CHECKLIST

■ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

■ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs (SDG 4.4 and 8.5).

■ The proposed solution targets a functional, deployed application rather than a theoretical business case study — live at https://sdc2026-lumina-nusantara-65cnugvuxdle5dgp9ky3fs.streamlit.app.

■ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon? **No.** All code, design and research was created during the SDC 2026 hackathon period.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create? **Yes** — declared below.

Name	Type (code/dataset/model)	License
Streamlit	code	Apache 2.0
pandas	code	BSD 3-Clause
O*NET database (U.S. Dept. of Labor)	dataset	CC BY 4.0 (attributed in-app)
WEF Future of Jobs indicators	dataset	Public report, cited
HRD Corp course directory	dataset	Public directory, cited

Name	Type (code/dataset/model)	License
all-MiniLM-L6-v2 (sentence-transformers)	model	Apache 2.0
RunPod Python SDK	code	Apache 2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon? **No.**

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature: _____

Name: DU YULIN

Date: _____ / _____ / 2026